Systems Requirement Specification

Smart Home Security System (SHSS)

VERSION	1.0
DATE	5 November 2026
AUTHOR	Ty
CONTENT STANDARD	ISO/IEC/IEEE 29148:2018, Clause 9.5
REQUIREMENTS	113
STATUS	Baselined for design

Document control

Document	System Requirements Specification (SysRS) — Smart Home Security System
Version	1.0
Date	5 November 2026
Author	Ty
Status	Baselined for design
Content standard	ISO/IEC/IEEE 29148:2018, Clause 9.5 — System requirements specification (SyRS) content
Section ordering	As set out in the project instructions and the supplied document template
Requirements	113 across 23 areas
Interfaces	26 specified in Section 6

> **A note on the clause number.** The project instructions direct the author to ISO/IEC/IEEE 29148:2018 “section 9.4. System requirements specification”. In the 2018 edition, Clause 9.4 is *Stakeholder requirements specification (StRS) content*; SyRS content is **Clause 9.5**, running 9.5.1 SyRS overview to 9.5.19 Assumptions and dependencies. This document follows the section ordering given in the project instructions and in the supplied template, and cites the standard by its correct clause.

Terms used in the requirement statements

shall marks a binding requirement. It is the only binding modal used; *should*, *may* and *will* do not appear in any requirement statement. Each requirement states exactly one obligation, so that it has exactly one verdict. **Unknown** is a reportable zone state distinct from *Clear*, and the distinction is load-bearing: a zone whose sensor is not reporting has not been measured, and reporting it as Clear would make an unmeasured zone indistinguishable from a safe one.

Requirements index

The assessment criteria and the supplied document template use different words for some of the same content — most visibly, the criteria group usability, performance, interfaces, maintainability and reliability under the heading “non-functional requirements”, which the template does not use. The template's headings are reproduced here without alteration. This index is the map.

WHERE THE ASSESSMENT CRITERIA SAY	WHERE IT IS IN THIS DOCUMENT
Define the system context	2.1 System Context, Figure 1
System functions	2.2 System functions
User characteristics — at least 2 types: homeowner/admin, residents	2.3 User characteristics — 5 types, homeowner/administrator and resident among them
Functional requirements — motion detection	3, SHSS-F-001 to 004
Functional requirements — video surveillance	3, SHSS-F-009 to 014
Functional requirements — alarm system	3, SHSS-F-015 to 019
Functional requirements — central control unit	3, SHSS-F-020 to 026
Functional requirements — user authentication	3, SHSS-F-027 to 032

WHERE THE ASSESSMENT CRITERIA SAY	WHERE IT IS IN THIS DOCUMENT
Non-functional — usability requirements	4 Usability Requirements
Non-functional — performance requirements	5 Performance Requirements
Non-functional — system interfaces (hardware or software)	6 System Interfaces
Non-functional — maintainability and reliability requirements	7.2 Maintainability and 7.3 Reliability
System specifications document contains all sections on the project page	All of 1 to 10, headings reproduced from the course template without alteration

Reference documents

REFERENCE	TITLE AND RELEVANCE
ISO/IEC/IEEE 29148:2018	Systems and software engineering — Life cycle processes — Requirements engineering. System requirements specification content is Clause 9.5.
ISO/IEC/IEEE 15288:2023	Systems and software engineering — System life cycle processes.
ISO/IEC/IEEE 42010:2022	Software, systems and enterprise — Architecture description.
ISO 9241-11:2018	Ergonomics of human-system interaction — Usability: definitions and concepts. Source of the effectiveness, efficiency and satisfaction measures in Section 4.
NIST SP 800-160 Vol. 1 Rev. 1	Engineering Trustworthy Secure Systems. Basis for treating security as a property across the life cycle rather than as a section.
UL 1023 / UL 985	Household burglar alarm system units; household fire warning system units.
47 CFR Part 15	Radio frequency devices. Governs the 908.42 MHz and 2.4 GHz emissions in Section 8.3.
IEEE 802.3af	Power over Ethernet. Governs ICD-P-06 and ICD-P-08.
WCAG 2.2 level AA	Source of the 4.5:1 contrast threshold in SHSS-U-004.

Contents

	Document control, references and requirements index
1	Introduction
1.1	Purpose
1.2	Scope
2	Overview
2.1	System Context
2.2	System functions
2.3	User characteristics
3	Functional Requirements
4	Usability Requirements
5	Performance Requirements
6	System Interfaces
7	System Operations

7.1 Human system integration requirements
7.2 Maintainability
7.3 Reliability
8 System modes and states
8.1 Physical requirements
8.2 Adaptability requirements
8.3 Environmental conditions
8.4 System security
8.5 Information management
8.6 Policies and regulations
8.7 System life cycle sustainment
8.8 Packaging, handling, shipping and transportation
9 Verification
10 Assumption and dependencies

Introduction

1.1. Purpose

This document specifies the system requirements for the **Smart Home Security System (SHSS)**, a residential security system comprising motion, door/window and water leak sensors, cameras, an alarm annunciator, device controllers and a central control unit, with local and remote user interfaces.

The system is being developed because the residential security products available to the reference household each force a choice the household is not willing to make. Cloud-first products put continuous video of the interior of the home on somebody else's storage and stop working when the broadband link does. Local-only products keep the video at home and offer no way to see it from anywhere else, which is where the owner is when it matters. Professionally monitored systems solve both and tie the household to a multi-year contract with equipment it does not own.

The purpose of the SHSS is to provide intrusion, access and water-leak detection with video verification, in which:

- detection, recording, alarm sounding and logging are performed **locally** and continue through a broadband outage;
- continuously recorded video never leaves the dwelling, and only alarm clips and explicit shares are uploaded;
- remote access, push notification and optional professional monitoring are available over an outbound-only connection with no inbound port exposed;
- the system reports what it does not know, rather than presenting an unmeasured zone as a safe one.

This specification is the basis for architectural design, component selection, the verification and validation plan, and acceptance. It states what the system shall do and the conditions under which it shall do it. It does not state how any element is implemented, except where an implementation property is itself the requirement — for example, that privacy masking is applied on the camera before encoding rather than in the viewer.

1.2. Scope

a. The system to be produced

The system is the **Smart Home Security System (SHSS)**, version 1.0. The abbreviation SHSS is used throughout. The system comprises every element listed in Section 2.1 as internal to the system boundary. The Mobile Application, Cloud Relay Service and Monitoring Service are within scope as interfacing elements, specified at their boundary only.

b. The results of the needs analysis

Stakeholder needs were elicited from the reference household — two adult owners, three residents, one dog of 16 kg, a detached garage and a finished basement — through structured interviews, a walkthrough of the dwelling, and review of twelve months of a neighbourhood incident log. The problems the household stated, in their own order of priority:

1. **“We don't find out until we get home.”** The existing arrangement — a doorbell camera and a window sticker — produces a notification only for the front door, and only when the broadband link is up.
2. **“We stopped arming it.”** A previous system produced roughly two false alarms a month, triggered by the dog. It was abandoned within a year. The household's own account is that a system that cries wolf is worse than no system, because it is a system they believe is on.
3. **“We don't want our living room on somebody's server.”** The household is willing to accept cloud involvement for notification and for alarm clips, and not for continuous interior video.
4. **“The basement flooded and we found out four days later.”** Water damage, not intrusion, is the loss the household has actually suffered.

5. **“We need the kids to be able to use it.”** Residents must be able to arm and disarm without being able to reconfigure the system or export video.

What the system will do. Detect intrusion at the perimeter and in the interior; detect water at floor level; verify detections with video; sound locally; notify every subscribed user with the zone, the event type and the time; record and retain video locally; provide remote viewing and control; log every state change and every authentication attempt; and continue to detect, record, sound and log through a broadband outage and through a mains outage of up to eight hours.

What the system will not do. It will not detect fire, smoke or carbon monoxide — those remain the responsibility of separately listed detectors, with which the SHSS interfaces only to relay an alarm. It will not perform facial recognition or identify named individuals. It will not dispatch emergency services directly; dispatch is performed by the optional Monitoring Service under its own agreement. It will not control heating, lighting or any other non-security function except where a controller is used as part of a security response. It will not operate as a life-safety system and is not a substitute for one.

c. Application, benefits, objectives and goals

The SHSS is applied to a single-family dwelling of up to 280 square metres over three storeys including a basement, with a detached garage within 30 metres of the main structure. The reference installation is 16 sensors, 4 cameras, 6 controllers, 1 wall panel and 1 control unit.

OBJECTIVE	MEASURE	TARGET
The owner learns about an intrusion in time to act	Detection to notification, 95th percentile	≤ 10 s (SHSS-P-001)
The system stays armed because it is trusted	False intrusion alarms per installation	≤ 1 per 90 days (SHSS-R-005)
An outage is not a security outage	Local detect, record, sound and log with the relay unreachable	Fully retained (SHSS-F-026)
Household privacy survives the security system	Continuous interior video leaving the dwelling	None (SHSS-IM-001)
Everyone in the house can operate it	First-time users arming in Away mode within 60 s, unaided	≥ 90% (SHSS-U-001)
Property damage from water is caught early	Leak detection to alarm, any arming mode	≤ 5 s (SHSS-F-008)

Overview

2.1. System Context

The system boundary encloses the Control Unit, the Power Supply Unit, the Sensor Bus Regulator, the PoE Network Switch, all sensors, cameras, the Alarm Annunciator, controllers and the Wall Panel. Three elements sit outside the boundary and are specified only at their interface: the Mobile Application, which runs on a device the household owns; the Cloud Relay Service; and the optional Monitoring Service.

Five human elements interact with the system. They are specified in Section 2.3, and they appear on the diagram below as actors because a context diagram that shows only equipment has left out the part of the system that makes most of the mistakes.

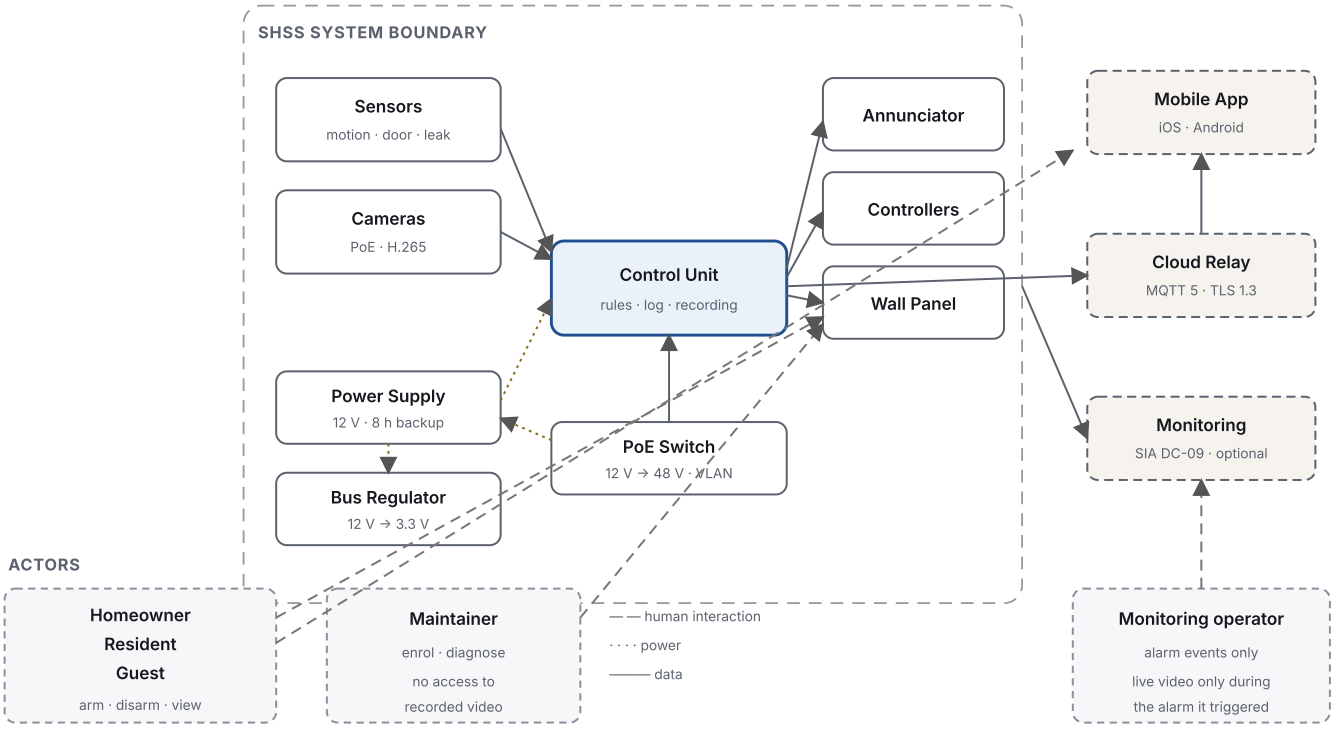


Figure 1 — System context. Solid lines carry data, dotted lines carry power, dashed lines are human interaction. Every line on this diagram has a row in the interface register at Section 6, and every row in that register has a line here.

2.2. System functions

The system's major capabilities, the conditions under which each applies, and the constraints on each.

CAPABILITY	WHAT IT DOES	CONDITIONS	CONSTRAINTS
`Intrusion detection`	Detects a person entering a monitored zone, at the perimeter and in the interior.	Active in Armed Stay (perimeter only) and Armed Away (all zones).	Interior zones are bypassed in Stay. Pet immunity to 18 kg.

CAPABILITY	WHAT IT DOES	CONDITIONS	CONSTRAINTS
Access monitoring	Detects the opening, closing and tampering of doors and windows.	Logged in every mode; alarms only when the zone is armed.	15 mm separation threshold.
Water leak detection	Detects water at floor level at enrolled probes.	Active in every mode including Disarmed.	Alarm within 5 s; not suppressible by arming state.
Video surveillance	Captures, records and streams video; classifies motion and persons on the camera.	Continuous local recording; clips uploaded only on alarm or share.	1080p15 minimum; privacy masks applied before encoding; 7–30 day local retention.
Alarm annunciation	Sounds interior and exterior annunciators and drives the strobe.	On any qualifying alarm, whether or not the relay is reachable.	85 dBA at 3 m; exterior capped at 10 minutes per event.
Notification	Delivers alarm and status notifications to subscribed users and to the Monitoring Service.	Immediate when the relay is reachable; queued and replayed when it is not.	10 s at the 95th percentile; zone, type and timestamp in every notification.
Arming and disarming	Changes the system between Disarmed, Armed Stay and Armed Away.	From the Wall Panel, the Mobile Application or a key fob.	Exit delay 30–120 s; open zones announced; duress code indistinguishable from a normal disarm.
Device control	Actuates locks, lights and relays as part of a security response.	On rule or on user command.	Never reports a commanded state as achieved without an acknowledgement.
Supervision	Monitors the liveness of every enrolled device and the health of every interface.	Continuous in every mode.	Two missed heartbeats raise a fault; a faulted zone reports Unknown, never Clear.
Event logging	Records every state change, command, alarm and authentication attempt.	Continuous, append-only, 365-day retention.	Every entry carries the authenticated identity that caused it.
User and access management	Authenticates users and enforces per-type permissions.	Before any function other than display of arming state at the panel.	Second factor for administrators; guest codes expire by time, not by revocation.
Degraded operation	Maintains local detection, recording, sounding and logging through loss of broadband, loss of mains, or a supervision fault.	Automatically, on detection of the loss.	8 h on backup battery with cameras unpowered; 7 days of event queueing; LTE failover carries alarm events only.

2.3. User characteristics

Five types of user, operator and maintainer. The count column is the number in each group for the reference installation; 29148 asks for it and it is routinely omitted, which is how systems end up with an account model that cannot express the household.

TYPE	NUMBER	LOCATION	DEVICES	NATURE OF USE
Homeowner / Administrator ADMIN	1 to 2 per installation	On premises and remote	iOS or Android app, web browser, wall-mounted touch panel	Full control. Arms and disarms in any mode, enrols and removes devices, creates and revokes user accounts, sets camera privacy zones and retention, views and exports recorded video, acknowledges and cancels alarms, authorises firmware updates.

TYPE	NUMBER	LOCATION	DEVICES	NATURE OF USE
Resident	2 to 6 per installation	On premises and remote	iOS or Android app, wall-mounted touch panel	Arms and disarms in Stay and Away modes, views live camera feeds from non-private zones, receives alerts, cancels an alarm they triggered. Cannot enrol devices, cannot change retention, cannot view or export recorded history older than 24 hours.
Guest	0 to 8, time-bounded	On premises	Wall-mounted touch panel, or app if invited	Disarms using a time-limited code during a window the homeowner sets. Receives no alerts, sees no cameras, and the code expires automatically rather than requiring revocation.
Maintainer ADMIN	1, a contracted installer	On premises, and remote with consent	Installer app with a separate credential	Enters Maintenance mode, replaces and re-enrols devices, runs diagnostics, reads logs. Cannot view recorded video, and every maintenance session is logged and notified to the homeowner.
Monitoring service operator	0 or 1, optional subscription	Off site, at the monitoring centre	Monitoring centre console over an authenticated API	Receives alarm events with location and zone, views live video only for the zone that triggered an alarm and only during the alarm, contacts the homeowner, dispatches emergency services. Holds no standing access to the system.

3. Functional Requirements

Functional requirements applicable to system operation, grouped by capability. Every requirement states one obligation and carries the rationale for its value, so that a later change can be evaluated against the reason rather than against the number alone.

ID	REQUIREMENT	RATIONALE	METHOD	TRACES TO
Motion detection				
SHSS-F-001	The Control Unit shall register a motion event from an enrolled motion sensor within 2 s of the sensor asserting detection.	2 s is the budget remaining for local processing once the 10 s end-to-end notification target (SHSS-P-001) has allowed 3 s for radio retry and 5 s for cloud fan-out.	Test	NEED-01
SHSS-F-002	Each motion sensor shall detect a human target crossing its field of view at between 0.3 m/s and 2.5 m/s at ranges up to 10 m.	0.3 m/s covers a deliberate slow approach, 2.5 m/s a running adult. 10 m covers the longest room in the reference installation.	Test	NEED-01
SHSS-F-003	Each motion sensor shall not assert detection for a quadruped of mass 18 kg or less moving below 1.0 m at ranges beyond 2 m.	Pet immunity to 18 kg covers the reference household's dog. Stated as a negative requirement because false alarms, not missed detections, are what cause a homeowner to stop arming the system.	Test	NEED-07
SHSS-F-004	The Control Unit shall bypass motion sensors designated interior while the system is in Armed Stay.	Occupants move freely at night; without this the Stay mode is unusable and gets replaced by not arming at all.	Demonstration	NEED-02
Door and window monitoring				
SHSS-F-005	The Control Unit shall register an opening event from an enrolled door or window sensor within 2 s of separation of the sensor and magnet exceeding 15 mm.	15 mm is below the gap a hand or tool needs, and above the movement a closed door makes in wind.	Test	NEED-03
SHSS-F-006	Each door and window sensor shall report a tamper event when its enclosure is opened or removed from its mounting surface.	A sensor an intruder can silently remove is a sensor that reports Clear while the opening is open.	Test	NEED-03
SHSS-F-007	The Control Unit shall classify each door and window zone as instant, delayed or perimeter-only, configurable per zone.	The front door needs an entry delay; a basement window does not, and giving it one hands an intruder 45 s.	Inspection	NEED-03
Water leak monitoring				
SHSS-F-008	The Control Unit shall raise a water leak alarm within 5 s of an enrolled leak sensor detecting continuity across its probes, in every arming mode including Disarmed.	Water damage does not wait for the system to be armed. Stating the mode independence here prevents it being implemented as an intrusion-class event.	Test	NEED-08
Video surveillance				
SHSS-F-009	Each camera shall capture video at a minimum of 1920x1080 pixels at 15 frames per second.	1080p15 is the lowest setting at which a face is identifiable at 4 m in the reference installation; frame rate matters less than resolution for identification.	Test	NEED-04
SHSS-F-010	Each camera shall produce a usable image at 0 lux using integrated infrared illumination at ranges up to 8 m.	Most intrusions occur in darkness; a camera that needs light records the events least likely to matter.	Test	NEED-04

ID	REQUIREMENT	RATIONALE	METHOD	TRACES TO
SHSS-F-011	The Control Unit shall begin recording the triggering camera within 1 s of an alarm event, including the 10 s preceding the trigger.	Pre-roll comes from a rolling buffer. Without it the recording starts after the thing worth seeing.	Test	NEED-04
SHSS-F-012	The Control Unit shall stream live video to an authenticated client with a glass-to-glass latency of no more than 3 s on the local network.	3 s is the point past which a user stops believing the feed is live and opens the door to look.	Test	NEED-05
SHSS-F-013	The Control Unit shall retain continuous recorded video on local storage for a configurable period of 7 to 30 days, defaulting to 14 days.	14 days covers a two-week absence. The upper bound is a storage constraint, the lower a usefulness one.	Inspection	NEED-04
SHSS-F-014	Each camera shall support configurable privacy masks that are applied on the camera before encoding.	Masking on the camera rather than in the viewer means the masked pixels are never recorded and cannot be recovered from the archive.	Demonstration	NEED-06
Alarm system				
SHSS-F-015	The Alarm Annunciator shall produce a sound pressure level of at least 85 dBA at 3 m within 1 s of being commanded.	85 dBA at 3 m is the common insurer and local-code expectation for a residential interior sounder.	Test	NEED-09
SHSS-F-016	The Control Unit shall sound the alarm locally when an alarm condition occurs, whether or not the Cloud Relay Service is reachable.	The single most important degradation property in the system: an internet outage must not be a way to disable the siren.	Test	NEED-09
SHSS-F-017	The Control Unit shall limit continuous exterior sounder operation to 10 minutes per alarm event.	Many municipal noise ordinances cap exterior sounding, commonly at 10 to 15 minutes; the interior sounder is not capped.	Test	CON-03
SHSS-F-018	The Control Unit shall accept a duress code that disarms the system, silences the annunciators and signals the Monitoring Service, with no indication at the user interface that it differs from a normal disarm.	A disarm under coercion has to look like a disarm to the person watching. The absence of a UI indication is the requirement.	Demonstration	NEED-10
SHSS-F-019	The Control Unit shall provide a configurable abort window of 15 to 60 s, defaulting to 30 s, during which a valid disarm cancels the signal to the Monitoring Service.	Most alarms are the owner's own mistake. An abort window is what keeps false dispatches, and the fines attached to them, down.	Test	NEED-07
Central control unit				
SHSS-F-020	The Control Unit shall maintain exactly one arming state at any instant, drawn from the set defined in Section 8.	Two components disagreeing about whether the system is armed is the defect that produces both missed alarms and false ones.	Analysis	NEED-11
SHSS-F-021	The Control Unit shall write every state change, command, alarm and authentication attempt to an append-only local event log.	Append-only so that an intruder with access cannot remove the record of having been there.	Inspection	NEED-12
SHSS-F-022	The Control Unit shall retain the event log for a minimum of 365 days.	Insurance claims and police reports routinely look back further than the video retention window.	Inspection	NEED-12
SHSS-F-023	The Control Unit shall support a minimum of 64 enrolled sensors, 16 cameras and 32 controllers concurrently.	Four times the reference installation, which is the growth the adaptability requirements assume.	Test	NEED-13
SHSS-F-024	The Control Unit shall raise a supervision fault when an enrolled device misses two consecutive heartbeat intervals.	Two intervals rather than one, so that a single lost radio frame does not produce a fault; the resulting detection time is stated per interface in Section 6.	Test	NEED-14

ID	REQUIREMENT	RATIONALE	METHOD	TRACES TO
SHSS-F-025	The Control Unit shall report a zone whose sensor is in supervision fault as Unknown rather than as Clear.	The distinction this whole specification turns on: an unmeasured zone is not a safe zone, and a user interface that shows green for both is worse than one that shows nothing.	Demonstration	NEED-14
SHSS-F-026	The Control Unit shall continue to detect, record, sound and log while the Cloud Relay Service is unreachable.	Local-first is the architectural decision taken in the server trade study; this is the requirement that holds the implementation to it.	Test	NEED-15
User authentication				
SHSS-F-027	The system shall authenticate every user before granting access to any function other than the display of current arming state at the Wall Panel.	Showing armed state on the panel without authentication is deliberate: it lets a resident confirm the state without a credential and reveals nothing an observer could not infer from the keypad.	Test	NEED-16
SHSS-F-028	The system shall require a second authentication factor for any account with administrator privilege.	The administrator account can disable the system and export video; a password alone is not proportionate to that.	Demonstration	NEED-16
SHSS-F-029	The system shall lock an account for 15 minutes after 5 consecutive failed authentication attempts within 10 minutes.	Rate limiting rather than permanent lockout, because a permanent lockout is itself a denial of service against the homeowner.	Test	NEED-16
SHSS-F-030	The system shall store user credentials using a memory-hard key derivation function with a per-credential salt.	Memory-hard rather than iterated-hash, because the threat is offline cracking of a stolen store on commodity GPUs.	Inspection	NEED-17
SHSS-F-031	The system shall expire a guest access code at the end of its configured window without requiring an explicit revocation.	Revocation that depends on someone remembering is revocation that does not happen.	Demonstration	NEED-18
SHSS-F-032	The system shall record the identity of the authenticated user against every arming, disarming and configuration change in the event log.	Attribution is what turns a log into evidence.	Inspection	NEED-12
Notification				
SHSS-F-033	The system shall deliver an alarm notification to every subscribed user device within 10 s of the alarm condition, measured at the 95th percentile over 100 consecutive events.	Stated as a percentile over a named sample size, because a mean latency hides the tail that users actually experience.	Test	NEED-01
SHSS-F-034	The system shall include the zone name, the event type and a timestamp in every alarm notification.	A notification that says only 'alarm' sends the owner home to find out which door.	Inspection	NEED-01
SHSS-F-035	The system shall queue notifications generated while the Cloud Relay Service is unreachable and deliver them on reconnection, each marked with its original event time.	Marked with the original time, or the owner reads a four-hour-old break-in as happening now.	Test	NEED-15
Arming and disarming				
SHSS-F-036	The system shall provide Armed Away, Armed Stay and Disarmed modes, selectable from the Wall Panel and the Mobile Application.	The minimum mode set a residential system is expected to have.	Demonstration	NEED-02
SHSS-F-037	The system shall provide a configurable exit delay of 30 to 120 s, defaulting to 60 s, on entry to Armed Away.	60 s is the time to leave by the front door and close it; the range covers a detached garage.	Test	NEED-02

ID	REQUIREMENT	RATIONALE	METHOD	TRACES TO
SHSS-F-038	The system shall announce any open perimeter zone at the start of the exit delay.	Arming with a window open is the commonest cause of an armed system that is not actually protecting anything.	Demonstration	NEED-02
SHSS-F-039	The system shall reflect a change of arming state on every connected user interface within 5 s.	Two residents acting on different views of the arming state is how a house ends up unarmed with everyone believing otherwise.	Test	NEED-11

4. Usability Requirements

Usability (quality in use) requirements. These are **non-functional** requirements. Each is stated as a measurable criterion of effectiveness, efficiency or satisfaction in a specific context of use, following ISO 9241-11; a usability requirement that cannot be measured on a panel of users is an opinion.

ID	REQUIREMENT	RATIONALE	METHOD	TRACES TO
SHSS-U-001	At least 90% of first-time resident users shall successfully arm the system in Away mode within 60 s of first picking up the Wall Panel, without consulting documentation.	Effectiveness, per ISO 9241-11. Measured on a panel of first-time users rather than asserted.	Test	NEED-19
SHSS-U-002	A trained resident shall complete a disarm from a locked phone screen in no more than 3 interactions.	Efficiency. Three is the budget before a resident starts leaving the system disarmed to avoid the friction.	Test	NEED-19
SHSS-U-003	At least 80% of residents shall rate the system's alert usefulness at 4 or higher on a 5-point scale after 30 days of use.	Satisfaction, and a proxy for false-alarm rate: an alert people stop reading is an alert that no longer works.	Test	NEED-07
SHSS-U-004	The Wall Panel shall present the current arming state in text and in a non-colour-dependent symbol, at a contrast ratio of at least 4.5:1.	Colour-only state coding fails for roughly 1 in 12 men. 4.5:1 is the WCAG 2.2 AA threshold for normal text.	Inspection	NEED-20
SHSS-U-005	Every alert and panel message shall be available in English and Spanish, selectable per user account.	A household where the system speaks a language a resident does not read has a resident who cannot respond to it.	Inspection	NEED-20

5. Performance Requirements

Critical performance conditions and the associated capabilities. These are **non-functional** requirements, covering (a) dynamic actions and rates, (b) endurance, design life and planned utilisation, and (c) performance in each operational phase and mode.

> **Where the end-to-end budget goes.** SHSS-P-001 allows 10 s from detection to a remote notification at the 95th percentile. That budget is spent as 2 s sensor-to-control-unit (SHSS-F-001), 3 s local processing and radio retry, and 5 s cloud fan-out. The component figures are derived from the budget rather than measured separately and then added up, which is the direction that produces a number nobody can meet.

ID	REQUIREMENT	RATIONALE	METHOD	TRACES TO
SHSS-P-001	The system shall deliver an intrusion alarm notification to a remote user device within 10 s of detection, at the 95th percentile, with the Cloud Relay Service reachable.	The end-to-end budget the component latencies are derived from: 2 s detection, 3 s radio and local processing, 5 s cloud fan-out.	Test	NEED-01
SHSS-P-002	The system shall process at least 20 simultaneous sensor events without loss and without any single event's notification exceeding the 10 s budget.	A forced entry trips several zones at once; a system that serialises and falls behind reports the break-in late.	Test	NEED-21
SHSS-P-003	The Control Unit shall sustain 8 concurrent 1080p video streams to local clients and 2 to remote clients without frame loss.	Sized for the reference installation's camera count plus the household's devices.	Test	NEED-05
SHSS-P-004	The system shall support at least 10 user sessions interacting concurrently with no interaction exceeding 2 s to first response.	Load testing target. Ten covers every resident device plus a monitoring console.	Test	NEED-21
SHSS-P-005	Each battery-powered sensor shall operate for at least 24 months on one set of cells under a duty cycle of 50 detection events and 2880 heartbeats per day.	Endurance, with the duty cycle stated. A battery life figure with no duty cycle attached is unverifiable.	Test	NEED-22
SHSS-P-006	The Control Unit shall operate for at least 8 hours on backup battery with all sensors supervised, cameras unpowered, and LTE failover active.	8 hours covers the overwhelming majority of residential outages. The exclusion of cameras is explicit because PoE is the dominant load and pretending otherwise would make the figure false.	Test	NEED-15
SHSS-P-007	The system shall have a minimum design life of 10 years, with a planned utilisation of continuous operation at 100% duty.	Life expectancy and utilisation rate, which 29148 asks for explicitly and which almost every specification omits.	Analysis	NEED-22

6. System Interfaces

These are **non-functional** requirements covering the interfaces among system elements, with the human element, and with external entities.

This section is the interface register for the system. Every interface carries its identifier, its two ends, the medium and protocol, the payload, the update rate, and — the column most interface documents leave out — **what the receiving element does when the interface is unavailable**.

That last column is a requirement, SHSS-I-001, rather than a courtesy. An interface that states a dependency without stating the behaviour on its absence has specified the happy path and called it an interface. The decision still gets made — by whoever writes each subsystem, differently each time, and usually discovered during integration.

Interface requirements

ID	REQUIREMENT	RATIONALE	METHOD	TRACES TO
SHSS-I-001	Every interface in Section 6 shall define the behaviour of the receiving element when the interface is unavailable.	An interface that states a dependency without stating the behaviour on its absence has described the happy path. The decision gets made anyway, by accident, differently in each subsystem.	Inspection	NEED-14
SHSS-I-002	All data interfaces crossing the system boundary shall use TLS 1.3 or later with certificate-based mutual authentication.	Mutual, not server-only: the Control Unit must also be unable to talk to an impostor relay.	Inspection	NEED-17
SHSS-I-003	All wireless sensor and controller interfaces shall use AES-128 encryption with Z-Wave S2 Authenticated pairing.	S2 Authenticated rather than S2 Unauthenticated, because the latter is vulnerable at the pairing step, which is exactly when an attacker would be present.	Inspection	NEED-17
SHSS-I-004	The system shall expose no inbound network port to the public internet.	Remote access is achieved by an outbound connection to the relay. An open inbound port on a home router is the single most common way consumer cameras are compromised.	Test	NEED-17
SHSS-I-005	The Control Unit shall interoperate with controllers conforming to Z-Wave or Matter over Thread.	Two open ecosystems rather than one proprietary one; the trade study in the controller matrix records why.	Demonstration	NEED-13

Interface register

ID	FROM → TO	TYPE	MEDIUM AND PROTOCOL	PAYLOAD	RATE	BEHAVIOUR WHEN UNAVAILABLE
ICD-P-01	Power Supply Unit → Control Unit	power	2-pin locking barrel, 18 AWG	12 V DC	—	Control Unit runs from the PSU backup battery; below 10.8 V it logs a shutdown event and halts cleanly.
ICD-P-02	Power Supply Unit → Sensor Bus Regulator	power	2-pin header, 22 AWG	12 V DC	—	Wired sensor bus unpowered; Control Unit reports all wired zones as Unknown, never as Clear.
ICD-P-03	Sensor Bus Regulator → Motion Sensor	power	JST-PH 2-pin (wired variant only)	3.3 V DC	—	Sensor falls back to its internal cells.
ICD-P-04	Power Supply Unit → Alarm Annunciator	power	Screw terminal, 16 AWG	12 V DC	—	Annunciator silent; Control Unit raises a supervision fault within 60 s.

ID	FROM → TO	TYPE	MEDIUM AND PROTOCOL	PAYLOAD	RATE	BEHAVIOUR WHEN UNAVAILABLE
ICD-P-05	Power Supply Unit → PoE Network Switch	power	2-pin locking barrel, 18 AWG	12 V DC	—	Switch unpowered; all PoE elements go offline together, which the Control Unit reports as a single infrastructure fault rather than as seven device faults.
ICD-P-06	PoE Network Switch → Camera	power	IEEE 802.3af PoE, RJ45, Cat 6	48 V DC	—	Camera offline; Control Unit marks the zone unmonitored and notifies the homeowner.
ICD-P-08	PoE Network Switch → Wall Panel	power	IEEE 802.3af PoE, RJ45, Cat 6	48 V DC	—	Panel dark; arming remains available from the app.
ICD-P-07	Power Supply Unit → Controller	power	Device mains or internal battery, per device	12 V DC	—	Controller unresponsive; last commanded state is retained by the device, not assumed by the system.
ICD-D-01	Motion Sensor → Control Unit	data	Z-Wave 908.42 MHz, AES-128, S2 Authenticated	json/sensor-event-v1	event-driven, plus a 30 s supervision heartbeat	Two missed heartbeats (60 s) raise a supervision fault and the zone reports Unknown.
ICD-D-02	Door/Window Sensor → Control Unit	data	Z-Wave 908.42 MHz, AES-128, S2 Authenticated	json/sensor-event-v1	event-driven, plus a 30 s supervision heartbeat	Two missed heartbeats (60 s) raise a supervision fault and the zone reports Unknown.
ICD-D-03	Water Leak Sensor → Control Unit	data	Z-Wave 908.42 MHz, AES-128, S2 Authenticated	json/sensor-event-v1	event-driven, plus a 60 s supervision heartbeat	Two missed heartbeats (120 s) raise a supervision fault and the zone reports Unknown.
ICD-D-16	PoE Network Switch → Control Unit	data	Ethernet 1000BASE-T, 802.1Q VLAN separating cameras from the household LAN	layer 2 frames	line rate	No camera or panel traffic reaches the Control Unit; reported as one infrastructure fault.
ICD-D-17	Control Unit → PoE Network Switch	data	SNMPv3 over Ethernet 1000BASE-T	json/switch-health-v1	0.0167 Hz (every 60 s)	Switch health unknown; port-level PoE faults are not distinguishable from device faults.
ICD-D-04	Camera → Control Unit	data	RTSP over TLS 1.3, Ethernet 1000BASE-T	H.265 video, json/vision-event-v1 metadata	continuous stream; events at up to 1 Hz	Control Unit marks the camera offline, notifies, and retains the last recorded segment.
ICD-D-05	Control Unit → Camera	data · cmd	ONVIF Profile S over TLS 1.3	json/camera-command-v1	on demand	Camera retains its current configuration; the Control Unit does not assume the command landed.
ICD-D-06	Control Unit → Alarm Annunciator	data · cmd	Supervised 12 V dry-contact pair, end-of-line resistor	binary/annunciate-v1	on demand, with a 1 Hz supervision poll	Open or short on the supervised pair raises a fault within 60 s; the system does not assume the sounder will work.
ICD-D-07	Alarm Annunciator → Control Unit	data	Supervised 12 V dry-contact pair, end-of-line resistor	binary/annunciate-fault-v1	1 Hz	Treated identically to a reported fault.
ICD-D-08	Control Unit → Controller	data · cmd	Z-Wave 908.42 MHz, AES-128, S2 Authenticated	json/actuate-v1	on demand	Command is retried three times, then a fault is raised. The system never reports a lock as locked on the basis of an unacknowledged command.

ID	FROM → TO	TYPE	MEDIUM AND PROTOCOL	PAYLOAD	RATE	BEHAVIOUR WHEN UNAVAILABLE
ICD-D-09	Controller → Control Unit	data	Z-Wave 908.42 MHz, AES-128, S2 Authenticated	json/actuate-ack-v1	on state change, plus a 60 s heartbeat	Controller state reports as Unknown.
ICD-D-10	Wall Panel → Control Unit	data · cmd	HTTPS/TLS 1.3 over Ethernet, mutual certificate	json/ui-command-v1	on interaction	Panel shows Offline and refuses input rather than queueing commands that may apply later.
ICD-D-11	Control Unit → Wall Panel	data	HTTPS/TLS 1.3 over Ethernet, mutual certificate	json/ui-state-v1	1 Hz	Panel shows Offline; see ICD-D-10.
ICD-D-12	Control Unit → Cloud Relay Service	data	MQTT 5 over TLS 1.3, client certificate	json/event-v2, json/clip-upload-v1	event-driven, plus a 30 s keepalive	The Control Unit continues to arm, detect, record and sound locally. Events queue to local storage for up to 7 days and replay on reconnection. Remote app access and push notification are unavailable; LTE failover carries alarm events only.
ICD-D-13	Cloud Relay Service → Control Unit	data · cmd	MQTT 5 over TLS 1.3, client certificate	json/remote-command-v1, json/firmware-manifest-v1	on demand	No remote commands are accepted; local control is unaffected.
ICD-D-14	Mobile Application → Cloud Relay Service	data	HTTPS/TLS 1.3, OAuth 2.0 with PKCE	json/app-api-v1	on interaction	App shows last known state with its age, and never presents a stale state as current.
ICD-D-15	Control Unit → Monitoring Service	data	SIA DC-09 over TLS 1.3, with an LTE bearer as the secondary path	SIA DC-09 alarm event	on alarm, plus a 300 s supervision poll	The monitoring centre raises its own supervision failure and contacts the homeowner. Local sounding and local notification are unaffected.
ICD-H-01	Wall Panel → Homeowner / Administrator	physical	Capacitive touch, 7-inch display, audible feedback	—	—	Arming and disarming remain available from the app and from a physical key fob.

7. System Operations

7.1. Human system integration requirements

Constraints on the allocation of functions to people, and on the interactions between people and equipment. Four areas require concentrated human engineering attention because the effects of human error in them are particularly serious:

1. **Disarm under duress.** The system must give a coerced user a way to summon help, and must present no difference an observer could detect. SHSS-F-018 states the behaviour; SHSS-O-003 states the absence of any indication, separately, because both can be built correctly while the pair is violated.
2. **Zone and sensor disablement.** Disabling a zone makes the house look armed while it is not. SHSS-O-002 requires a deliberate two-step confirmation.
3. **Alarm cancellation and dispatch.** An unnecessary dispatch carries a fine and erodes the response to the next one. The abort window (SHSS-F-019) is the human-factors control on it.
4. **Maintenance mode.** It inhibits the alarm outputs, so a session the owner does not know about is an unmonitored house (SHSS-O-004).

ID	REQUIREMENT	RATIONALE	METHOD	TRACES TO
SHSS-O-001	The Wall Panel shall position the disarm entry area no less than 1.2 m and no more than 1.4 m above finished floor level.	Reachable seated and by most adults standing, consistent with accessible reach ranges.	Inspection	NEED-20
SHSS-O-002	The Wall Panel shall require a deliberate two-step confirmation before any action that disables a sensor or zone.	Disabling a zone by mistake is a silent failure: the house looks armed and is not.	Demonstration	NEED-11
SHSS-O-003	The system shall present no visual, audible or haptic difference between a normal disarm and a duress disarm.	The paired requirement to SHSS-F-018, written separately because it constrains the user interface rather than the alarm logic, and both can be built correctly while the pair is violated.	Demonstration	NEED-10
SHSS-O-004	The system shall notify the homeowner within 60 s of entry to and exit from Maintenance mode.	Maintenance inhibits the alarm outputs. A maintenance session nobody told the owner about is an unmonitored house.	Inspection	NEED-23

7.2. Maintainability

Quantitative maintainability requirements for the planned maintenance and support environment. These are **non-functional** requirements, covering time, rate, complexity, maintenance action indices and accessibility.

ID	REQUIREMENT	RATIONALE	METHOD	TRACES TO
SHSS-M-001	The mean time to repair a failed sensor by field replacement shall be no more than 15 minutes, excluding travel.	Replacement rather than repair; the design consequence is that sensors mount on a removable plate.	Analysis	NEED-22
SHSS-M-002	The maximum corrective downtime for the Control Unit shall be no more than 4 hours, excluding parts availability.	Four hours bounds the window in which the house has only local sounders and no logging.	Analysis	NEED-22
SHSS-M-003	Sensor battery replacement shall require neither tools nor re-enrolment of the device.	Re-enrolment after a battery change is the reason households end up with four dead sensors and an armed system that covers half the house.	Demonstration	NEED-22

ID	REQUIREMENT	RATIONALE	METHOD	TRACES TO
SHSS-M-004	Preventive maintenance shall require no more than 2 staff-hours per year for the reference installation.	Maintenance rate, per 29148's maintainability examples.	Analysis	NEED-22
SHSS-M-005	The Control Unit shall apply a signed firmware update and return to its previous arming state within 10 minutes.	Returning to the previous arming state is the point: an update that leaves the system disarmed has disarmed the house.	Test	NEED-23
SHSS-M-007	The Control Unit shall roll back to the previously installed firmware image automatically if an updated image fails to reach a ready state within 10 minutes of installation.	Split from SHSS-M-005 so each has its own verdict. A failed update on a security system otherwise leaves the house unprotected until somebody drives there.	Test	NEED-23
SHSS-M-006	Every field-replaceable unit shall be reachable without removing any other unit.	Accessibility, per 29148. It is a layout constraint that has to be stated before the enclosure is designed.	Inspection	NEED-22

7.3. Reliability

Reliability requirements in quantitative terms, with the conditions under which they are to be met. These are **non-functional** requirements.

Reliability apportionment

The system availability target of 99.9% monthly (SHSS-R-001) is apportioned across the elements whose failure removes a security function. Elements in series take the product of their availabilities; the budget below leaves 0.02% unallocated as margin.

ELEMENT	APPORTIONED UNAVAILABILITY	REASONING
Control Unit	0.030%	Single point of failure for logging and rules; the largest single share.
Power Supply Unit	0.020%	Backup battery covers mains loss up to 8 h, so only PSU hardware failure counts here.
Sensor population	0.015%	Apportioned across 16 sensors; loss of one is a partial, not total, loss of function.
PoE Network Switch	0.010%	Loss removes cameras and the wall panel; sensors and sounders are unaffected.
Alarm Annunciator	0.005%	Supervised continuously, so failure is detected rather than latent.
Unallocated margin	0.020%	Held against integration effects not visible in element figures.

ID	REQUIREMENT	RATIONALE	METHOD	TRACES TO
SHSS-R-001	The system shall achieve an operational availability of at least 99.9% measured monthly, excluding mains outages longer than the 8-hour backup period.	99.9% monthly is about 43 minutes. The exclusion is stated because without it the figure measures the electricity grid.	Analysis	NEED-24
SHSS-R-002	The Control Unit shall have a mean time between failures of at least 50,000 hours at 25 degrees C.	About 5.7 years, against a 10-year design life, so one replacement over the life is planned for rather than a surprise.	Analysis	NEED-24
SHSS-R-003	Each sensor shall have a mean time between failures of at least 87,600 hours at 25 degrees C.	10 years, so sensor replacement is driven by batteries rather than by failures.	Analysis	NEED-24
SHSS-R-004	The probability of a missed intrusion detection by an enrolled, unfaulted sensor shall be no greater than 0.001 per qualifying event.	The reliability figure that matters to the user, as distinct from component MTBF.	Test	NEED-01

ID	REQUIREMENT	RATIONALE	METHOD	TRACES TO
SHSS-R-005	The system shall generate no more than 1 false intrusion alarm per installation per 90 days.	The number that decides whether the system stays armed. Most residential systems are abandoned over false alarms, not over missed detections.	Test	NEED-07

8. System modes and states

The system exists in exactly one arming state at any instant (SHSS-F-020), drawn from the set below. Degraded is orthogonal to the others: the system can be Degraded while Armed Away, and entry to Alarm remains available from it.

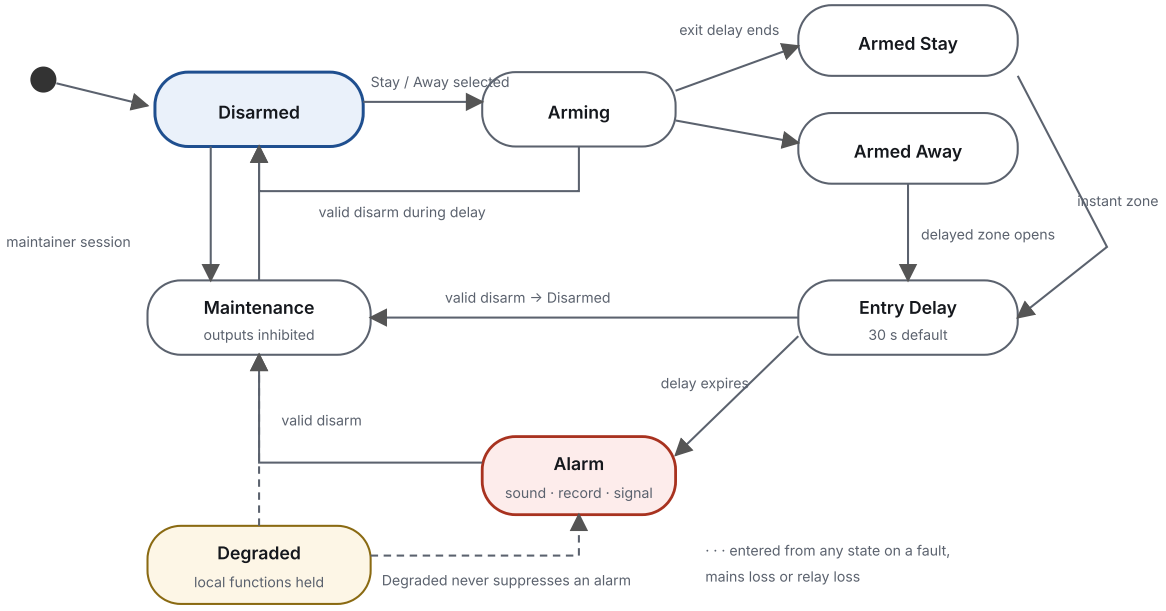


Figure 2 — Arming states and transitions. Degraded is drawn separately because it is orthogonal: the system can be Degraded while Armed Away, and the transition into Alarm remains available from it. A design in which Degraded suppressed alarms would be an outage that looks like a state.

MODE	WHAT IS TRUE IN IT	ENTERED BY	LEAVES TO
`Disarmed`	Sensors report and are logged. No intrusion alarm is raised. Water leak, smoke relay and tamper alarms remain active.	Power-on default, or a valid disarm from any armed mode	**Arming** — a user selects Stay or Away **Alarm** — a water leak or tamper event, which is armed-state independent **Maintenance** — an authenticated maintainer starts a session
`Arming`	Exit delay is running. Perimeter faults are announced so the user can correct them before the delay expires.	A user selects Stay or Away from Disarmed	**Armed Stay** — exit delay expires with Stay selected **Armed Away** — exit delay expires with Away selected **Disarmed** — a valid disarm during the delay
`Armed Stay`	Perimeter sensors are active. Interior motion sensors are bypassed so occupants can move without triggering an alarm.	Exit delay expires with Stay selected	**Entry Delay** — a delayed-zone opening **Alarm** — an instant-zone violation **Disarmed** — a valid disarm

MODE	WHAT IS TRUE IN IT	ENTERED BY	LEAVES TO
Armed Away	All intrusion sensors are active, interior motion included.	Exit delay expires with Away selected	**Entry Delay** — a delayed-zone opening **Alarm** — an instant-zone violation **Disarmed** — a valid disarm
Entry Delay	A delayed zone has opened and the user has a bounded window to disarm before the alarm sounds.	A delayed-zone opening while armed	**Disarmed** — a valid disarm before the delay expires **Alarm** — the delay expires
Alarm	Annunciators sound, the triggering zone records, notifications are sent, and the monitoring service is signalled if subscribed.	An instant-zone violation, entry delay expiry, duress code, water leak or tamper	**Disarmed** — a valid disarm, which also cancels the monitoring signal where the grace period allows
Maintenance	Alarm outputs are inhibited and every action is logged. The homeowner is notified on entry and on exit.	An authenticated maintainer starts a session, with homeowner consent	**Disarmed** — the session ends or its 4-hour limit expires
Degraded	A supervision fault, loss of mains, or loss of the cloud relay. The system keeps arming, detecting, recording and sounding locally, and states plainly which capability is unavailable.	Any supervision fault, mains loss, or relay loss	**Disarmed** — the fault clears and the user acknowledges **Alarm** — a qualifying event, which Degraded never suppresses

8.1. Physical requirements

Constraints on weight, volume and dimension, the construction characteristics of the installation, materials, nameplates and markings, interchangeability and workmanship.

ID	REQUIREMENT	RATIONALE	METHOD	TRACES TO
SHSS-PH-001	The Control Unit enclosure shall occupy no more than 300 x 250 x 90 mm.	Fits a standard utility-cupboard shelf.	Inspection	CON-01
SHSS-PH-007	The Control Unit shall weigh no more than 3.5 kg including its backup battery.	Split from SHSS-PH-001: volume and mass are verified by different measurements and can fail independently. 3.5 kg mounts on a single stud without a backing board.	Inspection	CON-01
SHSS-PH-002	The Control Unit enclosure shall be lockable by key or keyed-alike cam lock.	The control unit is the first thing an informed intruder attacks.	Inspection	NEED-17
SHSS-PH-008	The Control Unit shall report a tamper event within 2 s of its enclosure being opened.	Split from SHSS-PH-002: a lock is verified by inspection and a tamper report by test, and a locked enclosure that reports nothing is the more dangerous of the two failures.	Test	NEED-17
SHSS-PH-003	Exterior cameras and the exterior annunciator shall be rated to IP66 or better.	IP66 covers driving rain and dust, which is the residential exterior case.	Inspection	CON-02
SHSS-PH-004	All enclosure materials in occupied spaces shall meet UL 94 V-0 flammability.	A material requirement, which 29148 asks for explicitly under physical characteristics.	Inspection	CON-04
SHSS-PH-005	Every field-replaceable unit shall carry a permanent nameplate stating model, serial number, rated voltage and date of manufacture.	Nameplates and markings, per 29148. They are what makes a recall or a warranty claim possible years later.	Inspection	CON-04
SHSS-PH-006	Sensors of the same model shall be interchangeable without configuration beyond enrolment.	Interchangeability, per 29148. It is what lets a household keep one spare rather than one per zone.	Demonstration	NEED-22

8.2. Adaptability requirements

Requirements for growth, expansion, capability and contraction. Contraction is included deliberately: removing a zone, a device or a user without disturbing the rest is the half of adaptability that is normally left out and normally breaks.

ID	REQUIREMENT	RATIONALE	METHOD	TRACES TO
SHSS-A-001	The Control Unit shall accept additional sensors up to the capacity stated in SHSS-F-023 — 64 sensors, 16 cameras and 32 controllers — without hardware modification.	Growth within the stated capacity, not beyond it. The honest bound.	Demonstration	NEED-13
SHSS-A-002	The Control Unit shall provide at least one unused M.2 slot and one unused SATA bay for storage expansion.	The concrete form of 29148's own example: provision for growth designed in rather than promised.	Inspection	NEED-13
SHSS-A-003	The system shall allow a zone, a device or a user to be removed without affecting the operation of any other zone, device or user.	Contraction, which is the half of adaptability that gets left out.	Demonstration	NEED-13
SHSS-A-004	The Control Unit shall support the addition of a new device protocol by software update, with the radio hardware provided as a replaceable module.	Protocols turn over faster than control units. Matter over Thread did not exist when most installed panels shipped.	Inspection	NEED-13

8.3. Environmental conditions.

Natural, induced, electromagnetic-signal and self-induced environments, together with the legal and regulatory environment the system operates within.

ID	REQUIREMENT	RATIONALE	METHOD	TRACES TO
SHSS-E-001	Interior elements shall operate between 0 and 40 degrees C at 10% to 90% relative humidity, non-condensing.	Covers an unheated utility room in winter and an attic run in summer.	Test	CON-02
SHSS-E-002	Exterior elements shall operate between -30 and 60 degrees C at up to 100% relative humidity.	Covers the continental US climate range including direct solar gain on a south-facing wall.	Test	CON-02
SHSS-E-003	Wireless elements shall maintain the link budget required by Section 6 in the presence of 2.4 GHz and 5 GHz Wi-Fi at typical residential density.	The electromagnetic signal environment, and the reason the sensor radio is at 908.42 MHz rather than sharing the Wi-Fi bands.	Test	CON-05
SHSS-E-004	The system shall withstand a 4 kV contact electrostatic discharge to any user-accessible surface without loss of function.	Induced environment. A dry winter and a carpet are an EMC test nobody schedules.	Test	CON-02
SHSS-E-005	The system shall tolerate a mains voltage excursion of plus or minus 10% and a 20 ms interruption without entering Degraded.	Brownouts and transfer-switch gaps are routine and should not produce an alert.	Test	CON-02
SHSS-E-006	The system shall not transmit or receive outside the frequency allocations permitted by 47 CFR Part 15 in the deployment territory.	The legal half of the environmental section, which 29148 asks for and which is routinely omitted.	Inspection	CON-06

8.4. System security

Security requirements for both the facility that houses the system and the operational security of the system itself, including access limitations, data protection and recovery.

> **Security is a property of the whole system, not of this section.** Authentication requirements are in Section 3 because they are functions, interface encryption is in Section 6 because it constrains interfaces, and physical tamper detection is in Section 8.1 because it constrains the enclosure. This section carries what is left: protection of data at rest, integrity of what runs, and the security of the place the equipment is installed.

ID	REQUIREMENT	RATIONALE	METHOD	TRACES TO
SHSS-S-001	The system shall encrypt all recorded video at rest using AES-256.	The stolen-control-unit case: physical possession must not yield the archive.	Inspection	NEED-06
SHSS-S-002	The system shall verify a cryptographic signature on every firmware image before installing it.	Unsigned firmware on a device with cameras and door locks is the whole attack.	Test	NEED-17
SHSS-S-003	The system shall require physical presence at the Control Unit to perform a factory reset.	A remotely triggerable reset is a remotely triggerable disarm.	Demonstration	NEED-23
SHSS-S-004	The system shall apply least privilege such that no user type holds a permission its stated role does not require.	Stated as a property over the user table in Section 2.3, so it can be checked rather than asserted.	Inspection	NEED-16
SHSS-S-005	The facility housing the Control Unit shall be a lockable interior space not accessible from outside the dwelling.	29148 asks for the security of the facility that houses the system, not only of the system. For a home, that is where the box goes.	Inspection	NEED-17
SHSS-S-006	The system shall permit a homeowner to export and to permanently delete all personal data held about a named user within 30 days of request.	Data subject rights under CCPA/CPRA and comparable regimes; 30 days is the tighter of the common statutory windows.	Demonstration	CON-07
SHSS-S-007	The Control Unit shall restore its prior arming state and resume logging within 60 s of recovering from an unexpected restart.	Recovery, which 29148 lists alongside data protection. A system that comes back disarmed after a power blip has been disarmed by a power blip.	Test	NEED-23

8.5. Information management (Optional)

Requirements for the information the system receives, generates and exports, including what is retained, for how long, where it rests and what is backed up.

ID	REQUIREMENT	RATIONALE	METHOD	TRACES TO
SHSS-IM-001	The system shall store continuously recorded video only on local storage located inside the dwelling.	The data flow decision that follows from the server trade study: continuous video never leaves the house.	Inspection	NEED-06
SHSS-IM-005	The system shall upload a video clip to the Cloud Relay Service only for an alarm event or an explicit user share.	Split from SHSS-IM-001: one constrains where video rests, the other constrains when it moves, and a build can satisfy either while violating the other.	Inspection	NEED-06
SHSS-IM-002	The system shall retain event metadata for 365 days and alarm clips on the Cloud Relay Service for 90 days.	Two different retention periods because metadata is small and useful for far longer than video is.	Inspection	NEED-12
SHSS-IM-003	The system shall write an encrypted backup of configuration and event metadata, excluding video, to the Cloud Relay Service every 24 hours.	Archiving requirement. Excluding video is what keeps it small enough to be routine.	Test	NEED-12
SHSS-IM-004	The system shall restore a Control Unit from backup to a working configuration within 30 minutes.	A backup nobody has restored from is a hypothesis.	Test	NEED-22

8.6. Policies and regulations (Optional)

Regulatory requirements, organisational policies and the health and safety criteria basic to the design.

ID	REQUIREMENT	RATIONALE	METHOD	TRACES TO
SHSS-PR-001	The alarm annunciation subsystem shall comply with UL 985 for household fire warning interconnection and UL 1023 for household burglar alarm equipment, as applicable to its configuration.	Named because insurers and some jurisdictions require listed equipment, and because retrofitting a listing is not possible.	Inspection	CON-06
SHSS-PR-002	The system shall obtain and record explicit consent before enabling audio recording on any camera.	Several US states require all-party consent for audio recording. Video and audio sit on different legal footings and the system has to treat them differently.	Demonstration	CON-07
SHSS-PR-003	The system shall support configuration of the abort window and the exterior sounder duration to meet the local false-alarm ordinance in the deployment jurisdiction.	Municipal false-alarm ordinances vary and carry fines. Making this configurable is cheaper than a jurisdiction-specific build.	Inspection	CON-03
SHSS-PR-004	The system shall present no electromagnetic or acoustic hazard to occupants, with annunciator output not exceeding 110 dBA at 1 m.	Health and safety criteria, which 29148 asks for here. 110 dBA at 1 m bounds hearing risk for someone standing under it.	Inspection	CON-04

8.7. System life cycle sustainment (Optional)

Quality activities, measurement collection and analysis, spares, provisioning, documentation and training.

ID	REQUIREMENT	RATIONALE	METHOD	TRACES TO
SHSS-LC-001	The supplier shall provide security updates for the Control Unit for at least 7 years from the date of last sale.	An unsupported security device on a home network is worse than no device. Stating the window makes it a commitment rather than a hope.	Inspection	NEED-23
SHSS-LC-002	The supplier shall hold spare stock of every field-replaceable unit equal to at least 5% of the installed base per year for 7 years.	Sparing and provisioning, per 29148.	Inspection	NEED-22
SHSS-LC-003	The supplier shall provide installer training and a maintenance manual covering every field-replaceable unit before first shipment.	Support-personnel training and technical documentation, per 29148.	Inspection	NEED-22
SHSS-LC-004	The system shall report its own measurement set — uptime, event counts, false alarm count, battery levels and supervision faults — to the homeowner monthly.	Measurement collection, per 29148. It is also the only way the false alarm requirement SHSS-R-005 can be observed in the field.	Demonstration	NEED-24

8.8 Packaging, handling, shipping and transportation (Optional)

Requirements ensuring the system can be packaged, handled, shipped and transported within its intended operational context.

ID	REQUIREMENT	RATIONALE	METHOD	TRACES TO
SHSS-PK-001	Lithium cells shipped with or inside any element shall be packaged and labelled in accordance with UN 3480 or UN 3481 as applicable.	A real and commonly missed constraint: lithium batteries are regulated dangerous goods in air freight.	Inspection	CON-06

ID	REQUIREMENT	RATIONALE	METHOD	TRACES TO
SHSS-PK-002	Packaged elements shall withstand the drop and vibration profiles of ISTA 3A without loss of function.	ISTA 3A is the parcel-delivery profile, which is how these ship.	Test	CON-06
SHSS-PK-003	Packaged elements shall withstand storage and transport between -40 and 70 degrees C.	A delivery truck in July exceeds the operating range; the storage range has to be wider.	Test	CON-02

9. Verification

Each requirement in this specification is assigned exactly one verification method, drawn from the four methods below. The assignment is recorded against the requirement itself in Sections 3 to 8 and is collected into the matrix at the end of this section.

METHOD	WHAT IT MEANS HERE	USED WHEN
Test	Exercise the system or element under controlled, recorded conditions and measure the result against the stated value.	The requirement names a number.
Demonstration	Operate the system and observe that the required behaviour occurs. No instrumentation and no measurement.	The requirement is about a capability existing or a behaviour being absent, not about a quantity.
Inspection	Examine the item — drawing, configuration, enclosure, marking, source — without operating it.	Conformance is visible: a marking, a material, a retention setting, an algorithm choice.
Analysis	Establish conformance by modelling, calculation, or similarity to a previously qualified item.	Direct test is impossible, destructive or disproportionate — a 10-year design life, an MTBF figure, an availability target.

> **Review is not a verification method.** Design reviews, requirement reviews and readiness reviews are gates at which evidence is presented and a decision to proceed is taken. They are scheduled in the project plan and they do not close a requirement: a requirement marked verified because a meeting happened has not been verified. The four methods above are the only values the Method column takes.

Validation is separate from the above and answers a different question — not whether the system meets this specification, but whether this specification described the right system. It is performed by user acceptance testing against the needs recorded in Section 1.2b, by a pilot period of 30 days in the reference installation, and by load scenarios that reproduce the conditions under which the household actually uses the system. Validation activities begin before implementation, with review of this document against the stated needs, and do not wait for verification to complete.

Verification matrix

ID	AREA	METHOD	VERIFICATION EVIDENCE
SHSS-F-001	Motion detection	Test	TC-VER-001
SHSS-F-002	Motion detection	Test	TC-VER-001
SHSS-F-003	Motion detection	Test	TC-VER-001
SHSS-F-004	Motion detection	Demonstration	TC-VAL-004
SHSS-F-005	Door and window monitoring	Test	TC-VER-002
SHSS-F-006	Door and window monitoring	Test	TC-VER-002
SHSS-F-007	Door and window monitoring	Inspection	TC-VER-002
SHSS-F-008	Water leak monitoring	Test	TC-VER-003
SHSS-F-009	Video surveillance	Test	TC-VER-004
SHSS-F-010	Video surveillance	Test	TC-VER-004
SHSS-F-011	Video surveillance	Test	TC-VER-004, TC-VAL-001
SHSS-F-012	Video surveillance	Test	TC-VER-004, TC-VAL-003
SHSS-F-013	Video surveillance	Inspection	TC-VER-004
SHSS-F-014	Video surveillance	Demonstration	TC-VER-004

ID	AREA	METHOD	VERIFICATION EVIDENCE
SHSS-F-015	Alarm system	Test	TC-VAL-001
SHSS-F-016	Alarm system	Test	TC-VAL-001, TC-VAL-006
SHSS-F-017	Alarm system	Test	TC-VAL-001
SHSS-F-018	Alarm system	Demonstration	TC-VAL-004
SHSS-F-019	Alarm system	Test	TC-VAL-001
SHSS-F-020	Central control unit	Analysis	TC-VAL-004
SHSS-F-021	Central control unit	Inspection	TC-VER-005, TC-VER-006
SHSS-F-022	Central control unit	Inspection	Document inspection at design review
SHSS-F-023	Central control unit	Test	TC-VAL-005
SHSS-F-024	Central control unit	Test	TC-VER-001, TC-VER-002, TC-VER-003
SHSS-F-025	Central control unit	Demonstration	TC-VER-001, TC-VER-002
SHSS-F-026	Central control unit	Test	TC-VAL-006
SHSS-F-027	User authentication	Test	TC-VER-005, TC-VER-006
SHSS-F-028	User authentication	Demonstration	TC-VER-005
SHSS-F-029	User authentication	Test	TC-VER-006
SHSS-F-030	User authentication	Inspection	TC-VER-005
SHSS-F-031	User authentication	Demonstration	Document inspection at design review
SHSS-F-032	User authentication	Inspection	TC-VER-005
SHSS-F-033	Notification	Test	TC-VAL-001, TC-VAL-002
SHSS-F-034	Notification	Inspection	TC-VAL-001, TC-VAL-002
SHSS-F-035	Notification	Test	TC-VAL-006
SHSS-F-036	Arming and disarming	Demonstration	TC-VAL-004
SHSS-F-037	Arming and disarming	Test	TC-VAL-004
SHSS-F-038	Arming and disarming	Demonstration	TC-VAL-004
SHSS-F-039	Arming and disarming	Test	TC-VAL-004, TC-VAL-006
SHSS-U-001	Usability	Test	TC-VAL-004
SHSS-U-002	Usability	Test	TC-VAL-004
SHSS-U-003	Usability	Test	TC-VAL-002
SHSS-U-004	Usability	Inspection	TC-VAL-004
SHSS-U-005	Usability	Inspection	Document inspection at design review
SHSS-P-001	Performance	Test	TC-VAL-001, TC-VAL-002
SHSS-P-002	Performance	Test	TC-VAL-005
SHSS-P-003	Performance	Test	TC-VAL-007
SHSS-P-004	Performance	Test	TC-VAL-007

ID	AREA	METHOD	VERIFICATION EVIDENCE
SHSS-P-005	Performance	Test	Document inspection at design review
SHSS-P-006	Performance	Test	TC-VAL-006
SHSS-P-007	Performance	Analysis	Document inspection at design review
SHSS-I-001	Interfaces	Inspection	Document inspection at design review
SHSS-I-002	Interfaces	Inspection	TC-VER-005
SHSS-I-003	Interfaces	Inspection	Document inspection at design review
SHSS-I-004	Interfaces	Test	TC-VER-005
SHSS-I-005	Interfaces	Demonstration	Document inspection at design review
SHSS-O-001	Human system integration	Inspection	Document inspection at design review
SHSS-O-002	Human system integration	Demonstration	Document inspection at design review
SHSS-O-003	Human system integration	Demonstration	Document inspection at design review
SHSS-O-004	Human system integration	Inspection	Document inspection at design review
SHSS-M-001	Maintainability	Analysis	Document inspection at design review
SHSS-M-002	Maintainability	Analysis	Document inspection at design review
SHSS-M-003	Maintainability	Demonstration	Document inspection at design review
SHSS-M-004	Maintainability	Analysis	Document inspection at design review
SHSS-M-005	Maintainability	Test	Document inspection at design review
SHSS-M-007	Maintainability	Test	Document inspection at design review
SHSS-M-006	Maintainability	Inspection	Document inspection at design review
SHSS-R-001	Reliability	Analysis	Document inspection at design review
SHSS-R-002	Reliability	Analysis	Document inspection at design review
SHSS-R-003	Reliability	Analysis	Document inspection at design review
SHSS-R-004	Reliability	Test	TC-VAL-001
SHSS-R-005	Reliability	Test	Document inspection at design review
SHSS-PH-001	Physical	Inspection	Document inspection at design review
SHSS-PH-007	Physical	Inspection	Document inspection at design review
SHSS-PH-002	Physical	Inspection	Document inspection at design review
SHSS-PH-008	Physical	Test	Document inspection at design review
SHSS-PH-003	Physical	Inspection	Document inspection at design review
SHSS-PH-004	Physical	Inspection	Document inspection at design review
SHSS-PH-005	Physical	Inspection	Document inspection at design review
SHSS-PH-006	Physical	Demonstration	Document inspection at design review
SHSS-A-001	Adaptability	Demonstration	TC-VAL-005
SHSS-A-002	Adaptability	Inspection	Document inspection at design review

ID	AREA	METHOD	VERIFICATION EVIDENCE
SHSS-A-003	Adaptability	Demonstration	Document inspection at design review
SHSS-A-004	Adaptability	Inspection	Document inspection at design review
SHSS-E-001	Environmental	Test	Document inspection at design review
SHSS-E-002	Environmental	Test	Document inspection at design review
SHSS-E-003	Environmental	Test	Document inspection at design review
SHSS-E-004	Environmental	Test	Document inspection at design review
SHSS-E-005	Environmental	Test	Document inspection at design review
SHSS-E-006	Environmental	Inspection	Document inspection at design review
SHSS-S-001	Security	Inspection	Document inspection at design review
SHSS-S-002	Security	Test	Document inspection at design review
SHSS-S-003	Security	Demonstration	Document inspection at design review
SHSS-S-004	Security	Inspection	TC-VER-006
SHSS-S-005	Security	Inspection	Document inspection at design review
SHSS-S-006	Security	Demonstration	Document inspection at design review
SHSS-S-007	Security	Test	TC-VAL-006
SHSS-IM-001	Information management	Inspection	Document inspection at design review
SHSS-IM-005	Information management	Inspection	Document inspection at design review
SHSS-IM-002	Information management	Inspection	Document inspection at design review
SHSS-IM-003	Information management	Test	Document inspection at design review
SHSS-IM-004	Information management	Test	Document inspection at design review
SHSS-PR-001	Policy and regulation	Inspection	Document inspection at design review
SHSS-PR-002	Policy and regulation	Demonstration	Document inspection at design review
SHSS-PR-003	Policy and regulation	Inspection	Document inspection at design review
SHSS-PR-004	Policy and regulation	Inspection	Document inspection at design review
SHSS-LC-001	Life cycle sustainment	Inspection	Document inspection at design review
SHSS-LC-002	Life cycle sustainment	Inspection	Document inspection at design review
SHSS-LC-003	Life cycle sustainment	Inspection	Document inspection at design review
SHSS-LC-004	Life cycle sustainment	Demonstration	Document inspection at design review
SHSS-PK-001	Packaging and transport	Inspection	Document inspection at design review
SHSS-PK-002	Packaging and transport	Test	Document inspection at design review
SHSS-PK-003	Packaging and transport	Test	Document inspection at design review

Verification matrix — 113 requirements. Test 49 · Demonstration 19 · Inspection 37 · Analysis 8. Generated from the requirement set, so a requirement cannot be absent from it.

10. Assumption and dependencies

Assumptions are things taken to be true that have not been confirmed; dependencies are things outside the system or outside the supplier's control that the requirements rely on. Both are listed because a requirement derived from an unstated assumption cannot be re-derived when the assumption turns out to be false.

ID	ASSUMPTION OR DEPENDENCY	WHAT FOLLOWS IF IT DOES NOT HOLD
A-01	The dwelling has mains electricity at 120 V AC, 60 Hz, and a broadband connection with a public IPv4 or IPv6 address.	Section 8.3 states the tolerance to excursions; sustained absence of either moves the system to Degraded, which is specified, not assumed away.
A-02	The household owns smartphones capable of running a current iOS or Android release.	If not, the Wall Panel and a physical key fob remain sufficient for arming and disarming, and no requirement depends on a phone.
A-03	Z-Wave at 908.42 MHz is permitted in the deployment territory.	The territory is the United States. A different territory changes the radio module, which SHSS-A-004 makes replaceable for this reason.
A-04	The dwelling's internal construction does not include foil-backed insulation or steel studs in the walls between sensors and the Control Unit.	Either would degrade the 908 MHz link budget. A site survey at installation is the mitigation; a wired sensor bus is the fallback, which is why ICD-P-03 exists.
A-05	A resident is present to receive and act on a notification, or the Monitoring Service subscription is active.	The system detects and records; it does not respond. Without one of these, an alarm produces a recording and nothing else.
D-01	Dependency: the Cloud Relay Service is operated to an availability of at least 99.5% monthly.	Below that, SHSS-R-001 is still met because the availability target excludes remote access, which is the only capability the relay provides.
D-02	Dependency: the optional Monitoring Service holds the licences required to dispatch emergency services in the jurisdiction.	Outside the system boundary and outside the supplier's control. Named because the household's expectation of dispatch depends on it.
D-03	Dependency: lithium cells of the specified types remain commercially available for the 10-year design life.	CR123A and CR2032 are long-established formats. SHSS-LC-002 commits spare stock for 7 years, which does not cover the full design life and is stated here rather than left to be discovered.
D-04	Dependency: the local jurisdiction's false-alarm ordinance permits an abort window of at least 15 s.	SHSS-PR-003 makes the window configurable for this reason. A jurisdiction permitting none would require the monitoring signal to be sent immediately, raising the false-dispatch rate.
D-05	Dependency: UL listing of the annunciator subsystem is obtained before first shipment.	SHSS-PR-001. A listing cannot be retrofitted, and some insurers and jurisdictions require listed equipment.